

Few-Shot Prototype Head Adaptation for On-Device ECG Personalization on PSoC 6

Guilherme Silva, Pedro Silva, Gladston Moreira, and Eduardo Luz

Abstract—Wearable and bedside electrocardiogram (ECG) monitors must adapt to patient-specific morphology to maintain arrhythmia detection accuracy across users, yet personalization is typically performed offline and cannot account for individual physiology, electrode placement, or recording drift. On-device adaptation by backpropagation is expensive for microcontroller-class medical devices because backbone updates require backward passes and working memory. This letter proposes prototype-only head adaptation as a compact personalization primitive for TinyML ECG systems. A one-dimensional convolutional neural network (1-D CNN; 1,314 parameters and 72.6k multiply-accumulate operations per beat) is trained offline on the MIT-BIH Arrhythmia Database under an inter-patient protocol, frozen as a feature extractor, and exported to a PSoC 6 microcontroller. Patient-specific adaptation then reduces to computing closed-form class means in a 32-dimensional embedding space, requiring no convolutional backward pass, no iterative optimization, and only one forward pass per support beat. Prototype adaptation improves inter-patient macro-F1 from 0.635/0.639/0.646 to 0.731/0.771/0.797 at 1/5/10-shot, outperforming linear stochastic-gradient-descent (SGD) head fine-tuning at every shot count for the target tiny backbone. On-device replays at 1/5/10-shot match their host references (absolute prototype macro-F1 deviation < 0.00011), with 11.39 ms per segmented beat and a model-centric 5.2 KB flash / 22.2 KB SRAM footprint. A restricted variant that updates only the normal-class prototype from passively buffered sinus beats yields a consistent +0.05 macro-F1 gain, reducing the annotation burden during initial deployment.

Index Terms—ECG arrhythmia detection, TinyML, on-device learning, personalization, prototype networks, PSoC 6.

I. INTRODUCTION

PATIENT-SPECIFIC ECG morphology varies with physiology, electrode placement, and recording conditions, causing models trained on one population to degrade on unseen individuals under inter-patient evaluation protocols [1]. Wearable and bedside arrhythmia monitors, therefore, need a mechanism to personalize a deployed classifier for each new user without re-running a full training pipeline in the cloud. Tiny machine learning (TinyML) has made neural-network inference feasible on microcontroller-class IoT devices [2], [3], bringing the classifier physically close to the sensor, but the question of how to *adapt* that classifier on the same constrained device remains open.

Two barriers make on-device ECG personalization hard. The first is *computational*. Updating the backbone requires backward passes through convolutional layers and additional

working memory. Prior work has shown that on-device training is possible under 256 KB SRAM [4] and that 1-D CNN personalization for human activity recognition is feasible on STM32 hardware [5], yet both approaches rely on gradient-based updates costly relative to a medical device’s power and latency envelope. The second barrier is annotation. Since arrhythmic events are rare, a newly deployed monitor may accumulate many normal sinus beats before a labeled abnormal beat is available. Restricting adaptation to a linear head avoids convolutional backward passes but still requires iterative optimization. With normal-only class support, discriminative head fine-tuning risks biasing the classifier toward the normal class. Self-supervised and human-in-the-loop methods have highlighted this gap [6].

This letter addresses both barriers with a *prototype-only head adaptation* mechanism. A compact 1-D CNN (1,314 parameters, 72.6k multiply-accumulate operations – MACs – per beat) is trained offline on the MIT-BIH Arrhythmia Database [7] under the inter-patient protocol [1], and its convolutional backbone is then frozen and exported to a PSoC 6 microcontroller. Patient-specific adaptation reduces to computing closed-form class means in a 32-dimensional embedding space, one forward pass per support beat, no backward pass, no optimizer state. A restricted variant further reduces the annotation burden by updating only the normal-class prototype from passively buffered sinus beats, inheriting the abnormal prototype from offline training.

We verify the proposal through three complementary evaluations. (i) *Classification performance*: prototype adaptation improves inter-patient macro-F1 from 0.635/0.639/0.646 to 0.731/0.771/0.797 at 1/5/10-shot on eligible DS2 patients (vs. 0.664 pre-adaptation on all 22 DS2 records), outperforming a linear-SGD head baseline on a tiny neural-network backbone. The restricted normal-only variant yields a consistent absolute gain of approximately 0.05 in macro-F1 without any arrhythmia annotations. (ii) *On-device fidelity*: aggregate replay at 1/5/10-shot on the target Cortex-M4F matches the corresponding host macro-F1 within 0.00011, confirming that the closed-form update is numerically faithful on fp32 hardware. (iii) *Embedded cost*: Data Watchpoint and Trace (DWT) cycle counter profiling reports 11.39 ms per beat (2.8% of the 400 ms RR interval at 150 BPM), with a model-centric footprint of 5.2 KB flash and 22.2 KB SRAM on a device with 1 MB flash and 288 KB SRAM.

II. PROTOTYPE-ONLY HEAD ADAPTATION

Intuition. The deployed prototype classifier keeps one 32-D reference profile per class, its *prototype*. Calibrating to a new

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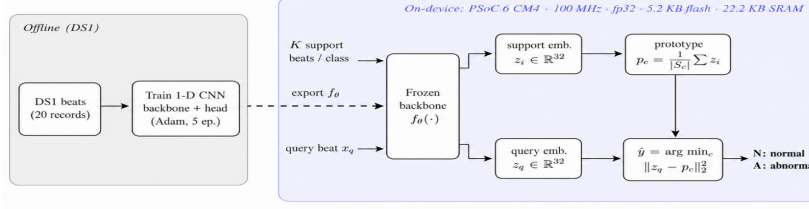


Fig. 1. Prototype-only ECG personalisation. *Offline*: a 1-D CNN is trained on DS1 and its frozen backbone is exported. *On-device*: support beats form 32-D class means; queries use nearest-prototype distance, with no backward pass, optimizer state, or backbone writes.

patient means passing a few labeled beats through the frozen feature extractor and averaging their feature vectors within each class. Each query beat is assigned to the nearest prototype. The prototypes are the head’s patient-specific parameters, computed in closed form without gradient descent during adaptation. Fig. 1 summarizes this flow.

Each ECG beat is a 200-sample window centered at an annotated R-peak. The backbone has two convolutional blocks (1→8→16) with kernel size 5, ReLU and max-pooling, followed by global average pooling and a projection to $D = 32$. It has 1,314 parameters and 72,576 MACs per beat. Upstream R-peak detection and segmentation costs and errors (missed/false peaks and window jitter) are excluded.

Let $f_\theta : \mathbb{R}^{200} \rightarrow \mathbb{R}^D$ be the frozen feature extractor. The patient-specific support set is $S = \{(x_i, y_i)\}_{i=1}^{KC}$, with K labeled beats per class and $y_i \in \{0, \dots, C-1\}$. Here, $C = 2$: normal ($c = 0$) and pooled abnormal ($c = 1$). Defining $I_c = \{i : y_i = c\}$ and $z_i = f_\theta(x_i)$, the class prototype is

$$p_c = \frac{1}{|I_c|} \sum_{i \in I_c} z_i, \quad |I_c| = K. \quad (1)$$

For a query embedding $z_q = f_\theta(x_q)$, the prediction is

$$\hat{y}(x_q) = \arg \max_c s_c(x_q), \quad s_c(x_q) = -\|z_q - p_c\|_2^2. \quad (2)$$

Expanding the score gives $s_c = -\|z_q\|_2^2 + 2p_c^\top z_q - \|p_c\|_2^2$. The first term is shared by all classes, so the decision is equivalent to a linear classifier *in embedding space*, with $w_c = 2p_c$ and $b_c = -\|p_c\|_2^2$ [8]. Thus, prototype adaptation replaces the DS1-trained linear head with a patient-specific head while keeping f_θ fixed.

The same class mean can be accumulated incrementally. Starting with $n_c = 0$ and $p_c = 0$, each new labeled support embedding z of class c updates

$$n_c \leftarrow n_c + 1, \quad p_c \leftarrow p_c + \frac{z - p_c}{n_c}. \quad (3)$$

Here, n_c includes the new example: the first sets $p_c = z$ and later examples move the mean by a fraction $1/n_c$ of their difference from it. Only the prototypes and counters need be retained. Their state and the query distance computation scale as $O(CD)$, excluding the frozen backbone. Multi-class use requires support for each rare class; a single mean may be insufficient for a class with several distinct feature clusters.

The closed-form mean is a deliberate choice for on-device adaptation. Diffusion-based prototype generation [9] and prototype-neighbor combinations [10] have been studied for

image classification. We do not compare their accuracy or deployment cost here, since they target different data and architectures.

For comparison, a linear head on the same frozen backbone is adapted using cross-entropy and 50 full-batch SGD updates (learning rate 0.05, zero momentum and weight decay), reusing support embeddings computed once.

Routine monitoring can provide many normal sinus beats before any patient-specific arrhythmic example becomes available. Requiring abnormal support would delay calibration despite this available normal data. We call this the *restricted (normal-only) adaptation* mode, in contrast to *full adaptation*, which computes both p_0 and p_1 from labeled patient support. In restricted mode, p_0 is computed from the patient’s normal support using Equation (1), while p_1 remains the mean abnormal embedding from DS1 training beats. Calibration can begin as soon as sufficient normal support is available, without waiting for target-patient arrhythmia annotations. This assumes normal support can be identified. Our evaluation uses reference annotations to select those beats.

III. PROTOCOL AND PSOC 6 IMPLEMENTATION

Experiments use the MIT-BIH Arrhythmia Database [7] under the inter-patient split of de Chazal *et al.* [1]. DS1 is used for training/validation and DS2 only for testing. The binary abnormal class pools distinct morphologies (LBBB, RBBB, PVC, APB); we therefore report macro-F1. For $K \in \{1, 5, 10\}$ support beats per class, a record must have at least $K + 1$ beats of each class. This excludes 4/7/8 of the 22 DS2 records, leaving 18/15/14 episodes.

Firmware validation targets the CY8CPROTO-063-BLE board, which integrates a PSoC 6 BLE device with a Cortex-M4F application core and 1 MB Flash / 288 KB SRAM [11]. The replay firmware stores exported weights and capped support/query episodes in flash, executes the frozen backbone and both head rules on the CM4, and logs metrics over UART. DWT cycle counts are collected from the CM4 Release build. The measured HF0 clock during profiling is 100 MHz.

Table I reports the model-centric memory budget for the deployable adaptation path. We report this separately from the full replay firmware image, which also contains board-support code, UART/profiling logic, the C runtime, and embedded replay assets. The source code to fully reproduce the work is available in <https://github.com/eduluzufop/TinyML-ECG-Prototype-Adaptation>.

TABLE I
MODEL-CENTRIC FOOTPRINT, NOT FULL FIRMWARE MEMORY. $K=10$
REPLAY CACHES 20 SUPPORT EMBEDDINGS; STREAMING PROTOTYPES
RETAIN 264 B OF ADAPTIVE STATE.

Item	Flash	SRAM
Backbone weights	4.9 KB	–
Head/prototypes	0.3 KB	0.26 KB
Static activation arrays	–	19.26 KB
Cached embeddings, $K = 10$	–	2.56 KB
Query embedding	–	0.13 KB
Subtotal	5.2 KB	22.2 KB
PSoC 6 budget	1024 KB	288 KB
Utilization	0.5%	7.7%

IV. RESULTS

Table II separates full eligible-DS2 accuracy from a matched host/device replay capped at 32 queries per patient. Prototype adaptation improves full-DS2 macro-F1 from 0.635/0.639/0.646 to 0.731/0.771/0.797 at 1/5/10-shot.

TABLE II
MEAN PER-PATIENT MACRO-F1 AND CM4 DWT ADAPTATION TIME.
REPLAY H/D IS HOST/DEVICE ON IDENTICAL CAPPED EPISODES (18/15/14
FOR $K=1/5/10$); DS2 USES EVERY QUERY OF EACH ELIGIBLE PATIENT.

Method	K	DS2	Replay H/D	Time
No adaptation	–	0.664 [†]	– [‡]	–
Linear SGD	1	0.671	0.689/0.689	25.97 ms
Prototype	1	0.731	0.798/0.798	22.83 ms
Linear SGD	5	0.704	0.732/0.732	124.52 ms
Prototype	5	0.771	0.785/0.785	114.04 ms
Linear SGD	10	0.709	0.682/0.682	247.69 ms
Prototype	10	0.797	0.804/0.804	228.06 ms

[†]All 22 DS2 records; shot-specific baselines are 0.635/0.639/0.646 after eligibility filtering. [‡]Replay pre-adaptation H/D varies with K : 0.611/0.611, 0.601/0.601, 0.632/0.632 at 1/5/10-shot.

Fig. 2 adds a non-gradient 1-nearest-neighbor (1-NN) classifier using all KC support embeddings on the same frozen embeddings. Its macro-F1 is 0.731/0.765/0.797, versus 0.731/0.771/0.797 for prototypes. The methods are identical at one shot, and at larger K , 1-NN retains KCD floats and comparisons versus CD for prototypes, without a consistent accuracy gain.

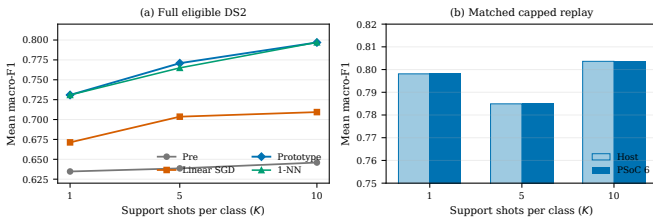


Fig. 2. Left: full eligible-DS2 mean macro-F1. Right: prototype host/device equivalence on the identical capped replay.

The replays contain 18/15/14 eligible episodes and 576/480/448 queries at 1/5/10-shot. Prototype host/device macro-F1 is 0.798114/0.798222, 0.784914/0.784933, and 0.803623/0.803643, and deviations are at most 0.000108. The

matched SGD deviations are below 0.0001 for every K after unifying its host and firmware update semantics.

The normal-only mode is weaker than full adaptation but potentially useful for initial calibration. At $K = 10$, it reaches 0.697 macro-F1, versus 0.646 before adaptation and 0.797 with full support. This suggests that normal morphology alone can improve the decision rule, though it cannot replace labeled abnormal support. Across shots, the paired gain is +0.043–0.052 macro-F1, roughly one-third of the full gain at $K = 10$, without target-patient arrhythmia annotations. Routine monitoring may supply abundant normal beats before abnormal events occur.

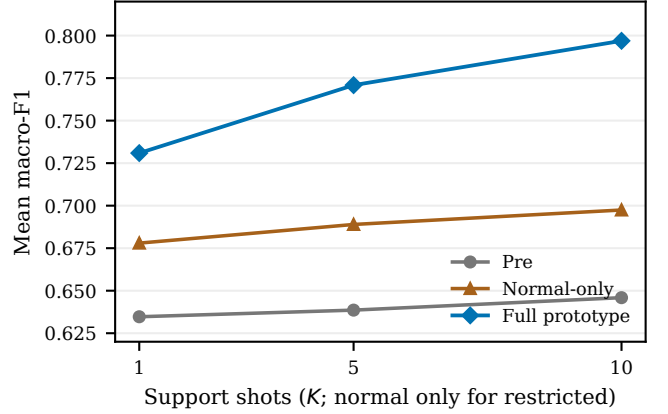


Fig. 3. Restricted (normal-only) vs. full adaptation. For each K , both use identical eligible DS2 records and query beats; the cohort can differ across K . Restricted updates p_0 only and retains DS1-derived p_1 .

Measured DWT profiling reports 1,139,141 cycles per already segmented beat for the tiny backbone, or 11.39 ms at 100 MHz (2.8% of a 400 ms RR interval at 150 BPM). Acquisition, R-peak detection, and segmentation were not profiled. Fig. 4 plots CM4 calibration time. Both methods embed each support beat once; at 10-shot, prototypes take 228.06 ms versus 247.69 ms for SGD, a 19.63 ms (7.9%) reduction. Calibration occurs at initialization or occasionally, for example after electrode repositioning or signal drift, rather than on every beat. Incoming signals can be buffered during calibration at an SRAM cost, but the prototype running mean does not require support embeddings to be retained.

Table III evaluates whether the prototype advantage is preserved under backbone scaling. The tiny backbone is both the primary embedded target and the setting where prototypes most consistently outperform SGD. Larger backbones narrow or reverse this advantage at low shot counts, indicating that the prototype-versus-SGD trade-off is capacity-dependent.

For the tiny backbone, class means and 1-NN are identical at $K = 1$; at $K = 5$ and 10, their mean macro-F1 differs by at most 0.006. Thus, storing every support embedding gives no consistent gain in these episodes. This does not imply that one centroid is always reliable, mislabeled or atypical support can shift the mean, while the pooled abnormal class may contain morphologies poorly represented by one center. The scaling study also limits generalization. Under fixed head settings, SGD leads at one shot for both larger backbones, but prototypes lead

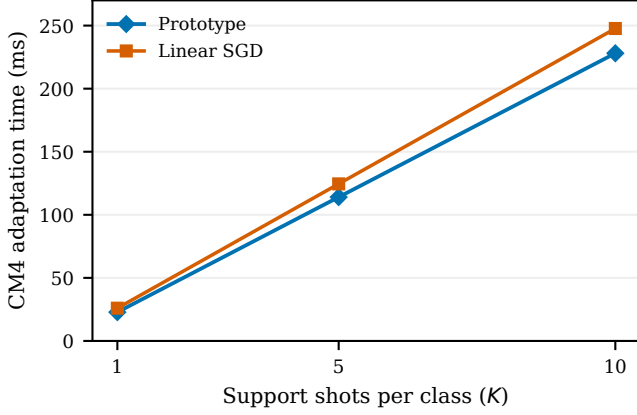


Fig. 4. PSoc 6 CM4 end-to-end calibration time: support embedding plus head update; query inference, replay control, and UART are excluded.

TABLE III

SCALING STUDY. FEW-SHOT COLUMNS USE FIXED HEAD SETTINGS ACROSS BACKBONES: SGD 0.05/50 UPDATES AND RAW EUCLIDEAN PROTOTYPES; NO DS2 RETUNING.

Variant	Params	MACs	Pre F1	1-shot	10-shot
Tiny	1,314	72.6k	0.664	Proto .731	Proto .797
Medium	2,618	157.0k	0.642	SGD .711	Proto .755
Large	4,930	274.2k	0.651	SGD .713	Proto .759

at ten shots (Table III). The preference for prototypes therefore pertains to the target tiny model and evaluated support sets.

V. DISCUSSION, LIMITATIONS, AND CONCLUSION

For the target tiny backbone, prototype adaptation matches the embedded constraints: a frozen feature extractor, no optimizer state, and a closed-form update in a 32-D embedding space. This complements prior on-device training [4] and HAR personalization [5]. Routine monitoring may supply abundant normal beats while labeled arrhythmic beats remain unavailable. Updating p_0 while retaining the DS1-derived p_1 permits initial personalization from this support. The paired $+0.043\text{--}0.052$ macro-F1 gain supports this role, though full adaptation performs better. The experiment assumes correctly identified normal beats.

The scaling study shows that larger backbones do not automatically improve inter-patient macro-F1, and they make the prototype-versus-SGD ranking shot-dependent. Therefore, our claim is not that prototypes dominate all personalization regimes, but that they are especially well matched to the target tiny model and MCU deployment path.

The DS1–DS2 split tests unseen patients but not other databases, sensors, or larger domain shifts. The task is binary rather than the AAMI five-class task, and upstream detection is excluded from the analysis. Firmware uses native-C `float32`. INT8 remains future work since quantization may perturb embedding geometry, prototype accumulation, and distance comparisons, so robustness must be measured rather than assumed [12].

In conclusion, prototype-only head adaptation improves inter-patient macro-F1 for the target tiny ECG model within the

reported PSoc 6 timing and model-centric memory footprint. Normal-only calibration offers a practical option for initial calibration when reliable normal support is available before patient-specific arrhythmia annotations. Its measured gain supports this use, with the abnormal prototype retained from DS1. Future work could validate automatic normal-support selection, extend the firmware to int8/TFLM, integrate upstream beat detection, and evaluate the full AAMI five-class setting.

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